5. Once the peptide bond is formed, the mRNA is pulled onward through the ribosome by exactly one codon. This shift allows the first, empty tRNA to drift out via the E site at the same time exposing the a new and subsequent codon to the A site.

6. The entire cycle repeats and every time a new aminoacid is added as per the information in the codons being exposed to the A site. Thus a polypeptide stats growing in the ribosome.

6.4.4.3 Termination of translation.

The growing polypeptide is stoped when the A site of the ribosome encounters one of the stop codons in the mRNA which are UAA, UAG, or UGA.

Stop codons are recognized by proteins called release factors, which fit neatly into the P site helping the synthesized polypeptide to be released.

The small and large ribosomal subunits separate from the mRNA and from each other and are free to take part in another different translation process.

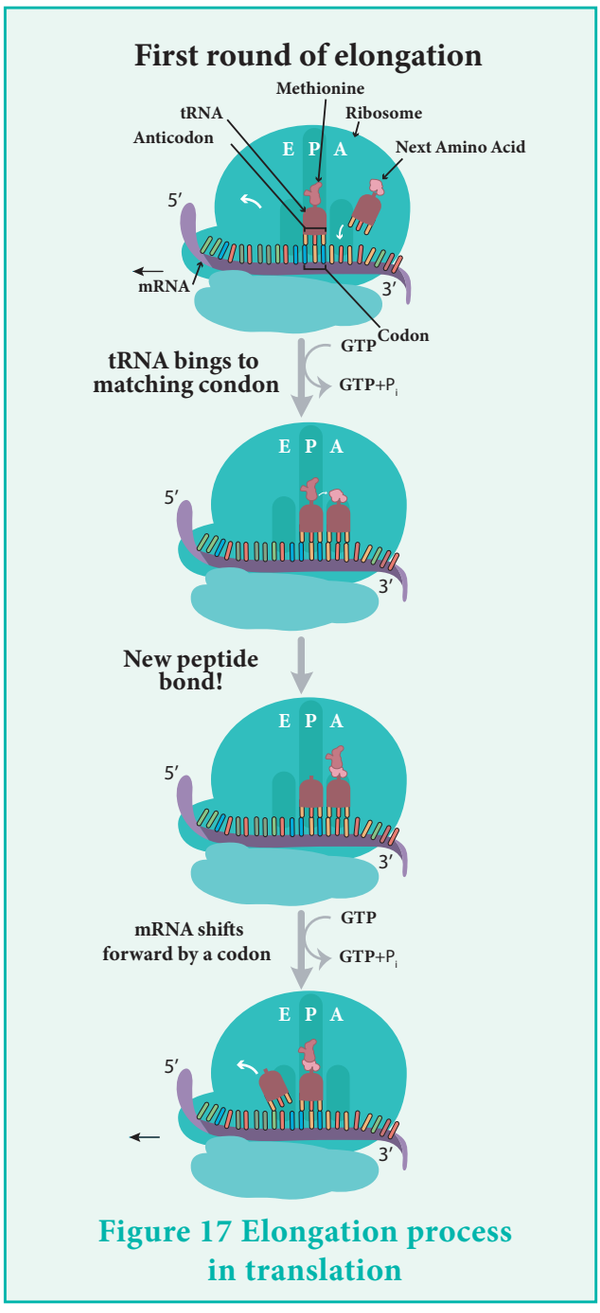


Figure 17 Elongation process in translation

6.4.5 Post-Translational Modifications

1. After translation, amino acids may be chemically altered or removed.

2. The chemical modifications include phosphorylation, glycosylation, glycation, nitrosylation, methylation, acetylation, lipidation etc

3. The new polypeptide will also fold into a distinct 3D structure and may join with other polypeptides to make a multi-subunit protein.

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